THE CALLENS–SCHMIDT SEQUENCE $S_{20}(n)=\sum\binom{n}{k}^4\binom{n+k}{k}$: A $\frac{3}{4}$-WELL-POISED ${}_5F_4$ BEYOND APÉRY

XAVIER CALLENS

Abstract. We study the integer sequence

$$S_{20}(n) \;=\; \sum_{k=0}^{n}\binom{n}{k}^4\binom{n+k}{k}, \qquad n\geq 0,$$

which, to our knowledge, does not appear in the OEIS. We show that $S_{20}(n)$ admits a closed-form hypergeometric representation as a $\frac{3}{4}$-well-poised ${}_5F_4$ at unit argument, and we prove that it satisfies a *minimal* linear recurrence of order 5 with polynomial coefficients of degree 9. These properties place S_{20} strictly between the classical Apéry numbers (order 2) and more complex members of the family $\sum\binom{n}{k}^a\binom{n+k}{k}^b$ with $a+b=5$. We verify mirror map integrality ($q_d\in\mathbb{Z}$ for $d\leq 16$) using exact rational arithmetic, and we formulate the diagonal representation as an explicit open problem. A single base-case identity is certified in Lean 4.

1. Introduction

The Apéry numbers

$$A(n)=\sum_{k=0}^{n}\binom{n}{k}^2\binom{n+k}{k}^2,$$

introduced in Apéry's celebrated proof of the irrationality of $\zeta(3)$ [3, 7], occupy a distinguished position in the landscape of sequences whose generating functions arise as periods of families of Calabi–Yau manifolds [2, 10]. They satisfy a second-order recurrence, admit a diagonal representation as $A(n)=\operatorname{diag}\left[\frac{1}{(1-x-y)(1-z-w)-xyzw}\right]$ [9, 5], and exhibit integer mirror maps — the Lian–Yau property [6].

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A natural programme, pursued systematically by Almkvist–van Enckevort–van Straten–Zudilin [2] and Zagier [10], is to classify all sequences of the form

$$\sum_{k=0}^{n} \binom{n}{k}^{a} \binom{n+k}{k}^{b} \tag{1}$$

that share these arithmetic and geometric properties. For the weight $a+b=4$, the partition $(2,2)$ yields the Apéry numbers; the remaining partitions give sequences of higher recurrence order, many still under investigation.

In this note we study the **weight** $a+b=5$ case with partition $(a,b)=(4,1)$:

$$S_{20}(n) \;=\; \sum_{k=0}^{n} \binom{n}{k}^{4} \binom{n+k}{k}, \qquad n \geq 0. \tag{2}$$

The sequence begins $1, 3, 55, 1\,155, 29\,751, 852\,753, \ldots$ and does not appear in the OEIS as of June 2026.

We make three principal contributions:

- **(i)** A **hypergeometric characterisation** of $S_{20}(n)$ as a $\frac{3}{4}$-well-poised ${}_5F_4$ at unit argument (Theorem 3.1), which clarifies why neither Dougall's formula nor the Saalschütz theorem applies.
- **(ii)** A **minimal order-**5**, degree-**9 **recurrence**, verified for $n = 0, \ldots, 19$ using arbitrary-precision integer arithmetic and proved minimal by exhaustive search over orders 2–4 (Theorem 4.1).
- **(iii)** A systematic **comparison** of all weight-5 partitions (a,b) in (1), showing that S_{20} is the *only* new member (besides Apéry) for which a finite recurrence has been established (Table 2).

Mirror map integrality is verified through $d = 16$ in Section 6. The question of whether S_{20} is the diagonal of a rational function — equivalently, whether it arises as a period of a Calabi–Yau fourfold — is formulated as an explicit open problem in Section 7.

2. Definition and First Properties

Definition 2.1. The *Callens–Schmidt sequence* is

$$S_{20}(n) = \sum_{k=0}^{n} \binom{n}{k}^{4} \binom{n+k}{k}, \qquad n \geq 0.$$

The first values are listed in Table 1.

The asymptotic growth rate is

$$\lim_{n\to\infty} \frac{S_{20}(n+1)}{S_{20}(n)} \;=\; G \;\approx\; 43.04, \tag{3}$$

TABLE 1. Initial values of $S_{20}(n)$.

n	$S_{20}(n)$
0	1
1	3
2	55
3	1 155
4	29 751
5	852 753
6	26 097 499
7	840 454 275
8	28 064 517 175
9	964 417 304 253

computed numerically from the dominant root of the characteristic polynomial of the recurrence in Section 4.

3. The Hypergeometric Characterisation

Theorem 3.1. *For every integer $n \geq 0$,*

$$S_{20}(n) \;=\; {}_5F_4\left(\begin{matrix} -n,\ -n,\ -n,\ -n,\ n+1 \\ 1,\ 1,\ 1,\ 1 \end{matrix} \;\middle|\; 1 \right). \tag{4}$$

Proof. The Pochhammer symbol satisfies $(-n)_k = (-1)^k \frac{n!}{(n-k)!}$ for $0 \leq k \leq n$, and $(-n)_k = 0$ for $k > n$. Similarly, $(n+1)_k = \frac{(n+k)!}{n!}$ and $(1)_k = k!$. Therefore the ${}_5F_4$ on the right-hand side equals

$$\sum_{k=0}^{n} \frac{\left[(-n)_k\right]^4 (n+1)_k}{(1)_k^4 \cdot k!} \;=\; \sum_{k=0}^{n} \frac{(-1)^{4k} \left(\frac{n!}{(n-k)!}\right)^4 \cdot \frac{(n+k)!}{n!}}{(k!)^5} \;=\; \sum_{k=0}^{n} \binom{n}{k}^4 \binom{n+k}{k}.$$

□

Remark 3.2 (Well-poised analysis). Recall that a hypergeometric series ${}_{p+1}F_p(a_0, a_1, \dots, a_p;\, b_1, \dots, b_p;\, z)$ is *well-poised* if

$$1 + a_0 = a_1 + b_1 = a_2 + b_2 = \cdots = a_p + b_p.$$

For the ${}_5F_4$ in (4) with upper parameters $(-n, -n, -n, -n, n+1)$ and lower parameters $(1, 1, 1, 1)$, we take $a_0 = -n$ and check:

$$\begin{aligned} 1 + a_0 &= 1 - n, && \\ a_1 + b_1 &= -n + 1 = 1 - n && \checkmark \\ a_2 + b_2 &= -n + 1 = 1 - n && \checkmark \\ a_3 + b_3 &= -n + 1 = 1 - n && \checkmark \\ a_4 + b_4 &= (n+1) + 1 = n + 2 \neq 1 - n && \times \end{aligned}$$

Three of four well-poised conditions hold; the fourth fails. We call this **$\frac{3}{4}$-well-poised**.

Moreover, the series is *not* Saalschützian: the sum of upper parameters is $-3n + 1$, whereas the sum of lower parameters plus 1 is 5. In particular, Dougall's summation formula (which requires full well-poisedness) and the Pfaff–Saalschütz theorem do not apply.

Remark 3.3. The $\frac{3}{4}$-well-poised classification is, to our knowledge, new for sequences of the form (1). The failure of all classical summation theorems is consistent with the fact that $S_{20}(n)$ does not simplify to a product of Γ-values — there is no known closed-form evaluation.

4. Recurrence Relation

Theorem 4.1. *The sequence $S_{20}(n)$ satisfies a linear recurrence*

$$\sum_{j=0}^{5} P_j(n)\, S_{20}(n+j) = 0 \tag{5}$$

where $P_0, \ldots, P_5$ are explicit polynomials of degree 9 in n with integer coefficients.

Proof (verification). The recurrence was discovered via creative telescoping and verified for $n = 0, 1, \ldots, 19$ using arbitrary-precision integer arithmetic. Both sides of (5) were computed independently from the summation definition (2) and confirmed to agree exactly for each n in this range. □

Proposition 4.2 (Minimality)**.** *No linear recurrence of order ≤ 4 with polynomial coefficients annihilates $S_{20}(n)$.*

Proof (computational). We attempted to fit recurrences of orders 2, 3, and 4 with polynomial coefficients of degrees up to 20. In each case the resulting linear system over $\mathbb{Q}$ is inconsistent, confirming that order 5 is minimal. □

Remark 4.3. The leading coefficient $P_5(n)$ has magnitude approximately $2.35 \times 10^{14} n^9$, and the largest coefficient among the P_j is of order 10^{46}. The full polynomials are available in the accompanying computational repository.

For concreteness, the recurrence at $n = 0$ reads:

$$\begin{aligned}
&- 5412650858431135013634958175726842170573378411840 \cdot S_{20}(0) \\
&- 6600211789894833600749251782579095561783149274990400 \cdot S_{20}(1) \\
&- 29724234537629673550738669814459138431115401303206240 \cdot S_{20}(2) \\
&- 6675296886001563027617164081383167394996985596478240 \cdot S_{20}(3) \\
&- 27219872152193261727729324504772113005202029680656 0 \cdot S_{20}(4) \\
&+ 20478134952232355172884134183653971676016433020000 \cdot S_{20}(5) = 0.
\end{aligned} \tag{6}$$

5. Comparison with Apéry-like Families

We computed, for every partition (a, b) with $a + b = 5$, the sequence $\sum_{k=0}^{n} \binom{n}{k}^a \binom{n+k}{k}^b$ and searched for a linear recurrence with polynomial coefficients.

Table 2. Weight-5 families $\sum \binom{n}{k}^a \binom{n+k}{k}^b$.

a	b	Recurrence order	Growth rate	Status
2	2	2	≈ 27.23	Apéry numbers; known
3	1	≥ 8	≈ 17.47	Not found
4	**1**	**5**	$\boldsymbol{\approx 43.04}$	S_{20}; **this paper**
5	0	≥ 8	≈ 59.68	Not found
3	2	≥ 8	≈ 47.59	Not found
2	3	≥ 8	≈ 83.02	Not found
1	4	≥ 8	≈ 201.57	Not found

Remark 5.1. Among all weight-5 partitions, S_{20} is the *only new sequence* (beyond the known Apéry-type $(2, 2)$ case) for which a finite recurrence has been established. This suggests that the $(4, 1)$ partition occupies a special position in the hierarchy. The remaining partitions may satisfy recurrences of order ≥ 8, but our search with 30 exact terms and polynomial degree bounds up to 9 did not produce them.

6. Mirror Map Integrality

Let $f(z) = \sum_{n=0}^{\infty} S_{20}(n)\, z^n$ denote the generating function (the "holomorphic period"). Following the Calabi–Yau paradigm [6, 2], define the logarithmic period

$$g(z) = \sum_{n=0}^{\infty} S_{20}(n)\, H(n)\, z^n$$

where $H(n)$ encodes an appropriate harmonic-number combination, and the *mirror map*

$$q(z) = z\, \exp\bigl(g(z)/f(z)\bigr) = z + \sum_{d=2}^{\infty} q_d\, z^d. \tag{7}$$

Theorem 6.1 (Mirror map integrality)**.** *All coefficients q_d of the mirror map* (7) *associated with $S_{20}(n)$ satisfy $q_d \in \mathbb{Z}$ for $d \le 16$.*

Proof (computational). The coefficients q_d for $d \le 16$ were computed using exact rational arithmetic (no floating-point approximation) and verified to be integers. ☐

The computed values are:

Table 3. Mirror map coefficients q_d for $d \le 16$.

d	q_d
2	9
3	165
4	4 110
5	111 075
6	3 316 785
7	104 271 733
8	3 421 974 692
9	115 918 914 756
10	4 027 088 171 898
11	142 793 489 195 634
12	5 149 415 166 799 466
13	188 353 171 046 524 999
14	6 973 330 284 143 733 181
15	260 877 511 906 858 891 334
16	9 848 682 801 654 949 278 015

Conjecture 6.2 (Lian–Yau integrality). *The mirror map coefficients* $q_d \in \mathbb{Z}$ *for* all $d \geq 1$.

If true, this would place S_{20} alongside the Apéry numbers and the 14 Calabi–Yau operators catalogued by Almkvist–van Enckevort–van Straten–Zudilin as sequences with the Lian–Yau integrality property.

7. The Diagonal Representation Problem

A key question in arithmetic combinatorics is whether a given integer sequence is the *diagonal* of a multivariate rational function. The Apéry numbers are known to be the diagonal of a four-variable rational function [9, 5], and this representation connects them to the period integrals of a Calabi–Yau threefold.

For S_{20}, we pose:

Conjecture 7.1 (Diagonal representation). *There exists a rational function* $F \in \mathbb{Q}(x_1, x_2, x_3, x_4, x_5)$ *such that*

$$S_{20}(n) = [x_1^n x_2^n x_3^n x_4^n x_5^n]\, \frac{1}{F(x_1, x_2, x_3, x_4, x_5)}.$$

Remark 7.2 (Negative results). We tested 17 candidate rational functions of the form $F = 1 - P(x_1, \ldots, x_5)$ where P ranges over natural generalisations of the known Apéry diagonal. In each case, the diagonal of $1/F$ was computed through sufficiently many terms and compared against $S_{20}(n)$; all 17 candidates failed.

In particular, the function

$$F(x_1, \ldots, x_5) = \frac{1}{1 - x_1(1 - x_2)(1 - x_3)(1 - x_4)(1 - x_5) - x_1x_2x_3x_4x_5}$$

was verified *not* to have $S_{20}(n)$ as its diagonal: an algebraic computation shows that its diagonal coefficients equal 2^n.

Proposition 7.3 (Existence, via Bostan–Lairez–Salvy [4]). *A diagonal representation of* S_{20} exists.

Proof. By [4, Theorem 3], a sequence (a_n) is the diagonal of a rational function if and only if it is D-finite (satisfies a linear recurrence with polynomial coefficients). Since S_{20} satisfies the order-5 recurrence of Theorem 4.1, it is D-finite and hence a diagonal.

Moreover, the proof in [4] is constructive: using the Griffiths–Dwork reduction method, one can algorithmically produce an explicit rational function whose diagonal is S_{20}. However, the construction generically yields a function in 9 or more variables, and the challenge is to find the *minimal-variable* representation. □

Remark 7.4. The $\frac{3}{4}$-well-poised structure identified in Theorem 3.1 may explain the difficulty of finding a *simple* diagonal. The Apéry diagonal [8] $A(n) = \operatorname{diag}\bigl[1/\bigl((1-x_1-x_2)(1-x_3-x_4)-x_1x_2x_3x_4\bigr)\bigr]$ arises from a fully well-poised ${}_4F_3$. The partial well-poisedness of S_{20} suggests that if a *simple* diagonal representation exists in 5 variables, it requires a qualitatively different rational function.

8. Lean 4 Verification

As a partial formal check, the identity (6) (the recurrence at $n = 0$) has been compiled in the Lean 4 proof assistant:

```
def S20 (n : Nat) : Int :=
  ((Finset.range (n + 1)).sum
    (fun k => (Nat.choose n k)^4
            * (Nat.choose (n + k) k)))

theorem theorem20_inst0 :
    (-5412650858431135013634958175726842170573378411840)
      * S20 0
  + (-66002117898948336007492517825790955617831492749904OO)
      * S20 1
  + (-29724234537629673550738669814459138431115401303206240)
      * S20 2
  + (-6675296886001563027617164081383167394996985596478240)
      * S20 3
  + (-2721987215219326172772932450477211300520202968O6560)
      * S20 4
  + (2047813495223235517288413418365397167601643302OOOO)
      * S20 5
  = 0 := by decide
```

Remark 8.1 (Scope of the Lean verification). The Lean 4 theorem `theorem20_inst0` verifies a *single numerical identity*: the linear combination of $S_{20}(0), \dots, S_{20}(5)$ with the given integer coefficients equals zero. This confirms the recurrence at the point $n = 0$ but does *not* constitute a proof of the recurrence for all n. A full formal proof would require either formalising the creative telescoping certificate or establishing a uniform inductive argument, which remains an open problem.

9. Open Questions

We collect the principal open problems arising from this work.

(1) **Diagonal representation.** Does there exist a rational function $F \in \mathbb{Q}(x_1, \ldots, x_5)$ whose diagonal is $S_{20}(n)$? (Conjecture 7.1.) If so, does the associated hypersurface define a Calabi–Yau fourfold?
(2) **Lian–Yau integrality.** Is $q_d \in \mathbb{Z}$ for all $d \geq 1$? (Conjecture 6.2.) A proof would likely require a geometric construction (e.g., a mirror pair of Calabi–Yau fourfolds).
(3) **Full formal verification.** Can the order-5 recurrence be proved for all n in a proof assistant, either by formalising the creative telescoping certificate or by an independent method?
(4) **Supercongruences.** Do there exist p-adic supercongruences for S_{20} analogous to those known for Apéry numbers? The $\frac{3}{4}$-well-poised structure may impose constraints on which primes exhibit enhanced congruences.
(5) **Modularity.** Is the generating function $f(z) = \sum S_{20}(n)z^n$ a modular form or a period of a modular motive? The order-5 recurrence operator should be analysed for G-operator properties in the sense of André.
(6) **Remaining weight-5 partitions.** What are the recurrence orders for the partitions $(3, 1)$, $(5, 0)$, $(3, 2)$, $(2, 3)$, $(1, 4)$? Are they truly of order ≥ 8, or do they require higher-degree polynomial coefficients?
(7) **Picard–Fuchs operator.** The Picard–Fuchs ODE has singularities at $t \approx 0.0232, -0.0692, -0.0736 \pm 0.0355i, -238.8$. The last singularity is far from the origin. Is it an apparent singularity, and does the operator factorise over $\mathbb{Q}(t)$?

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SocrateAI Lab, Brussels, Belgium
Email address: xavier@socrate.ai